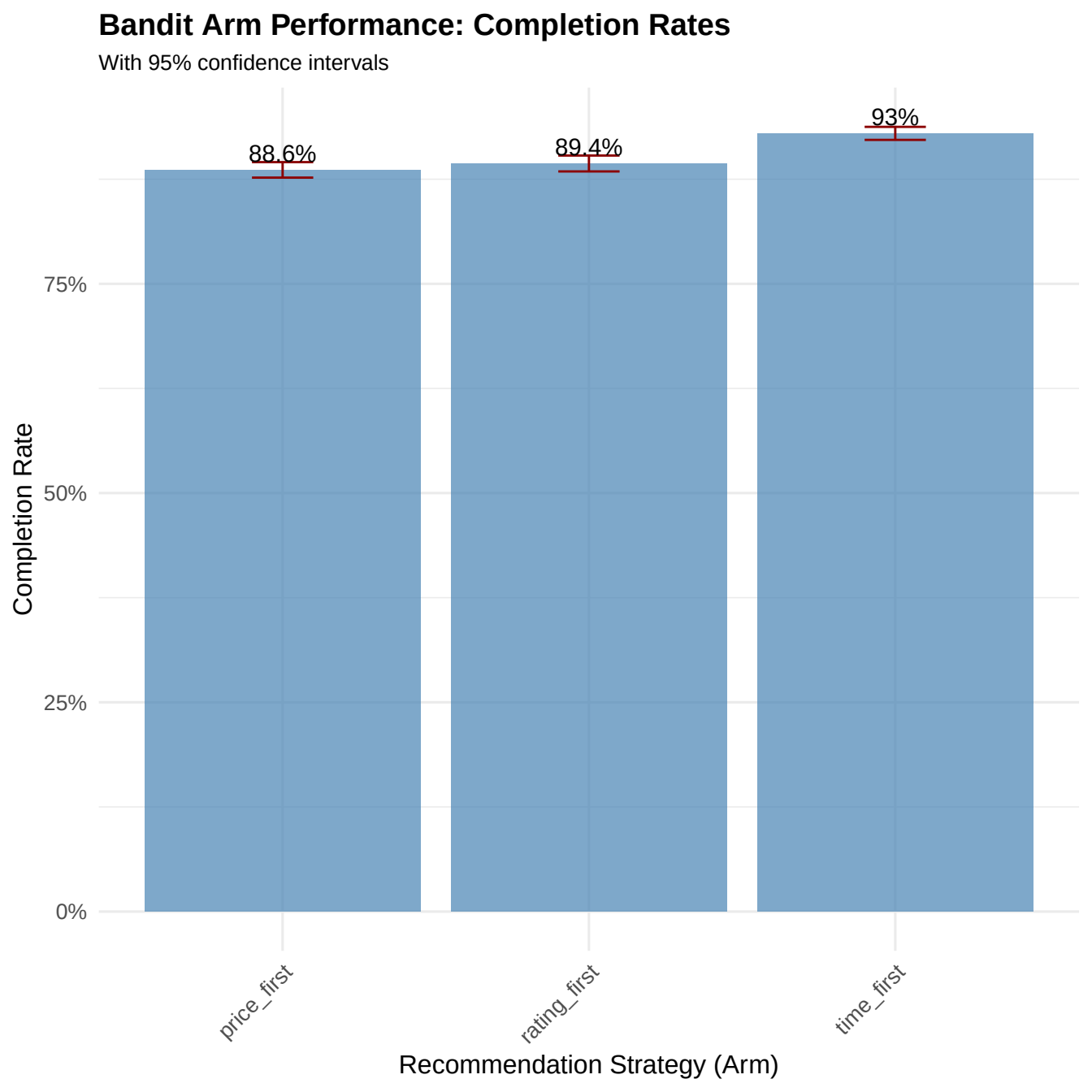
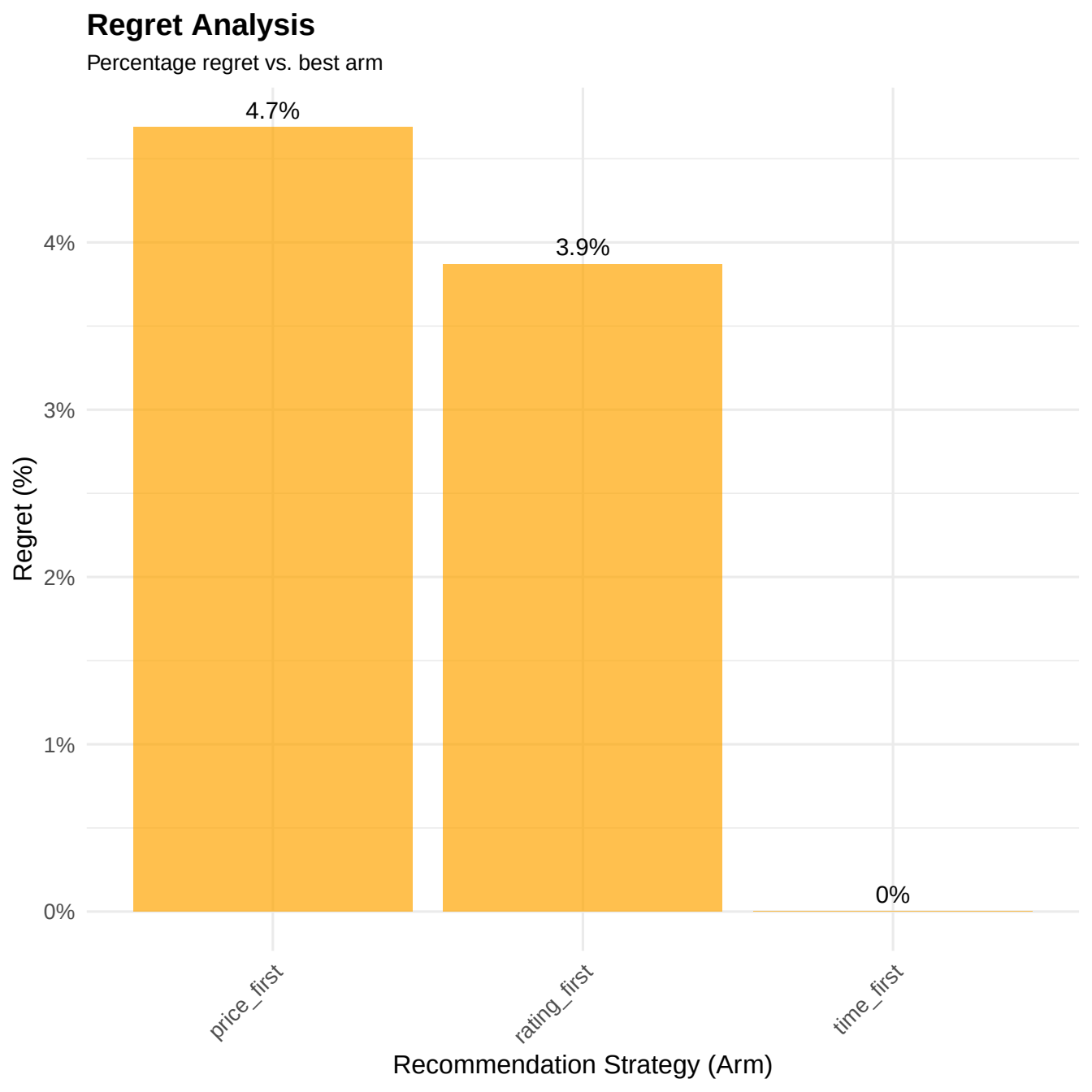
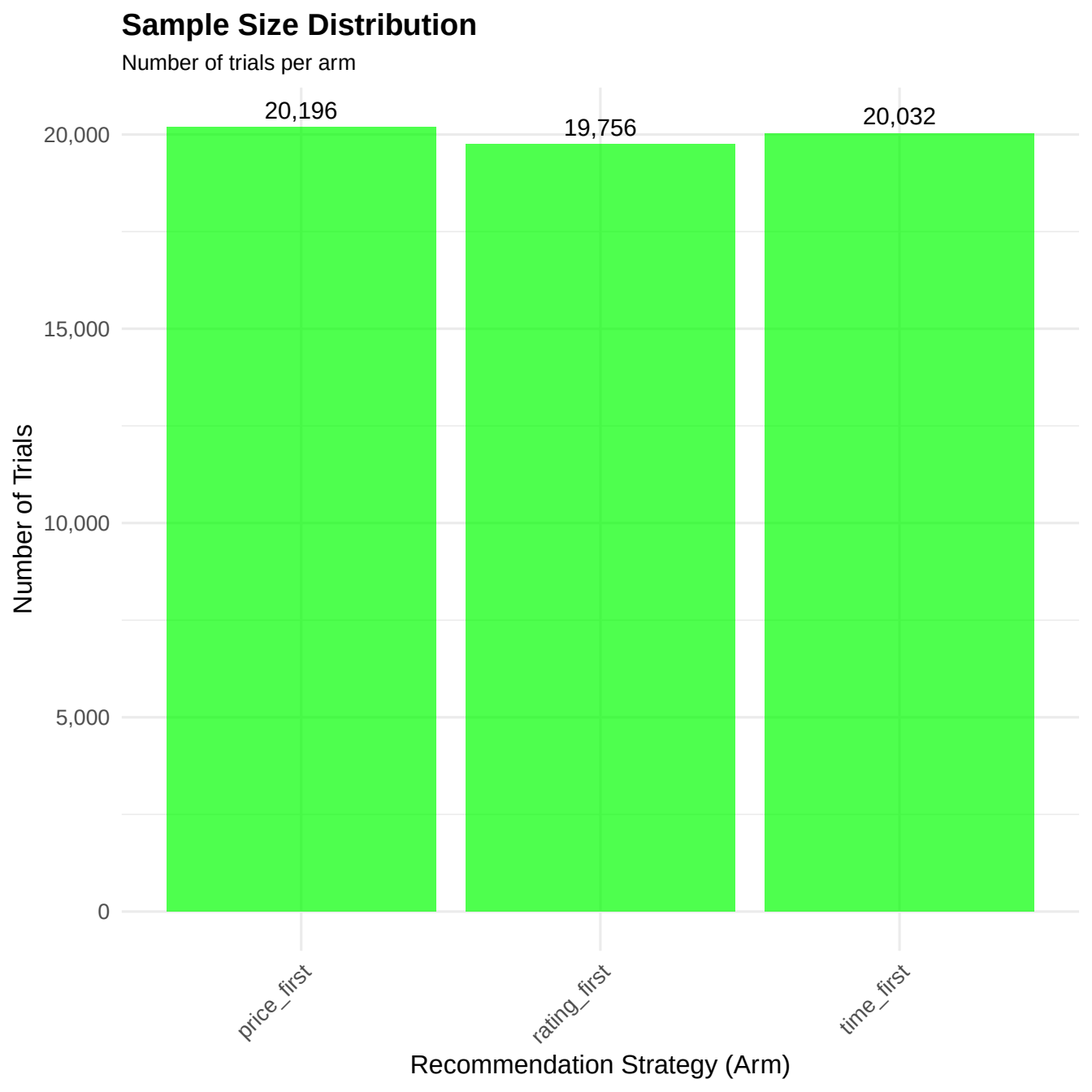
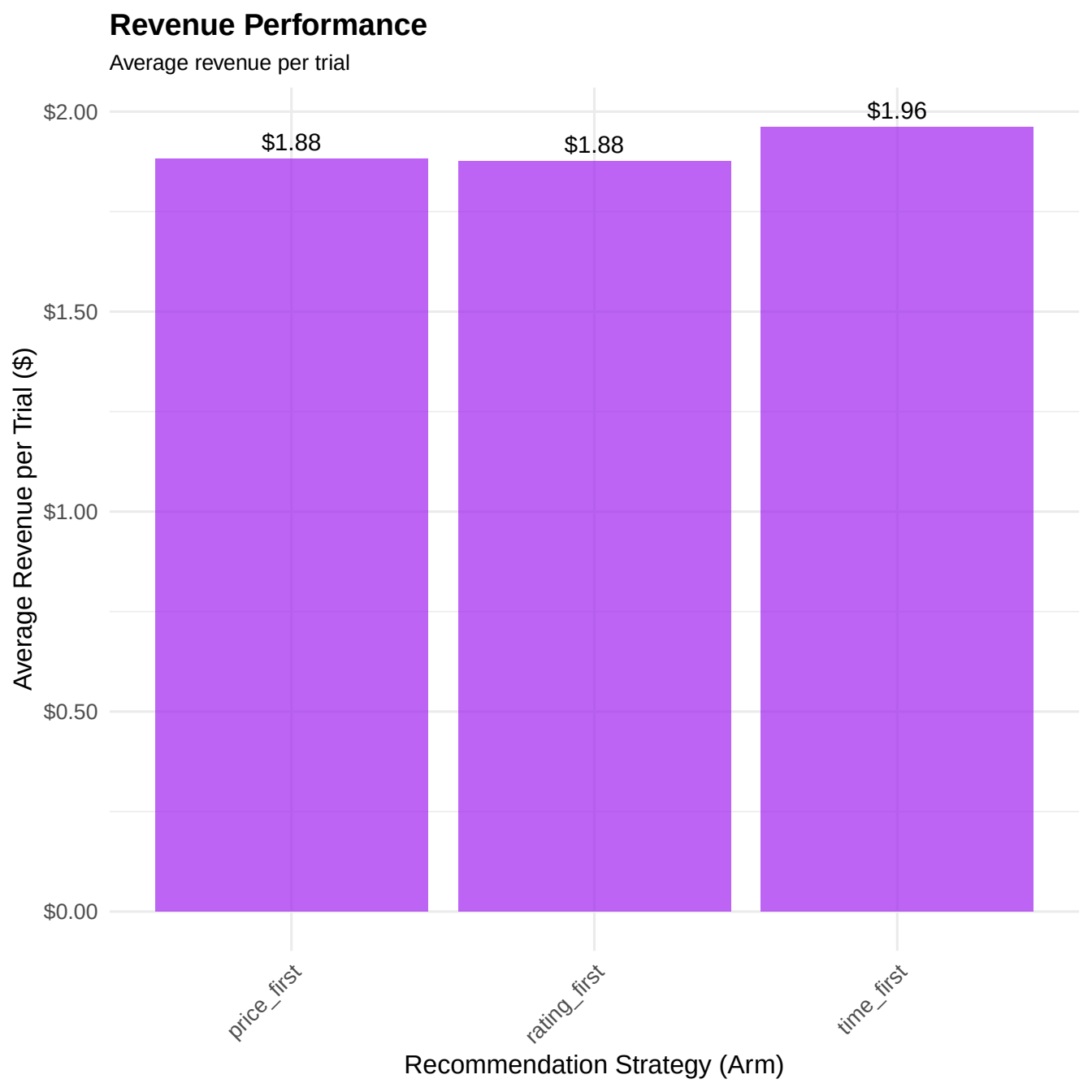
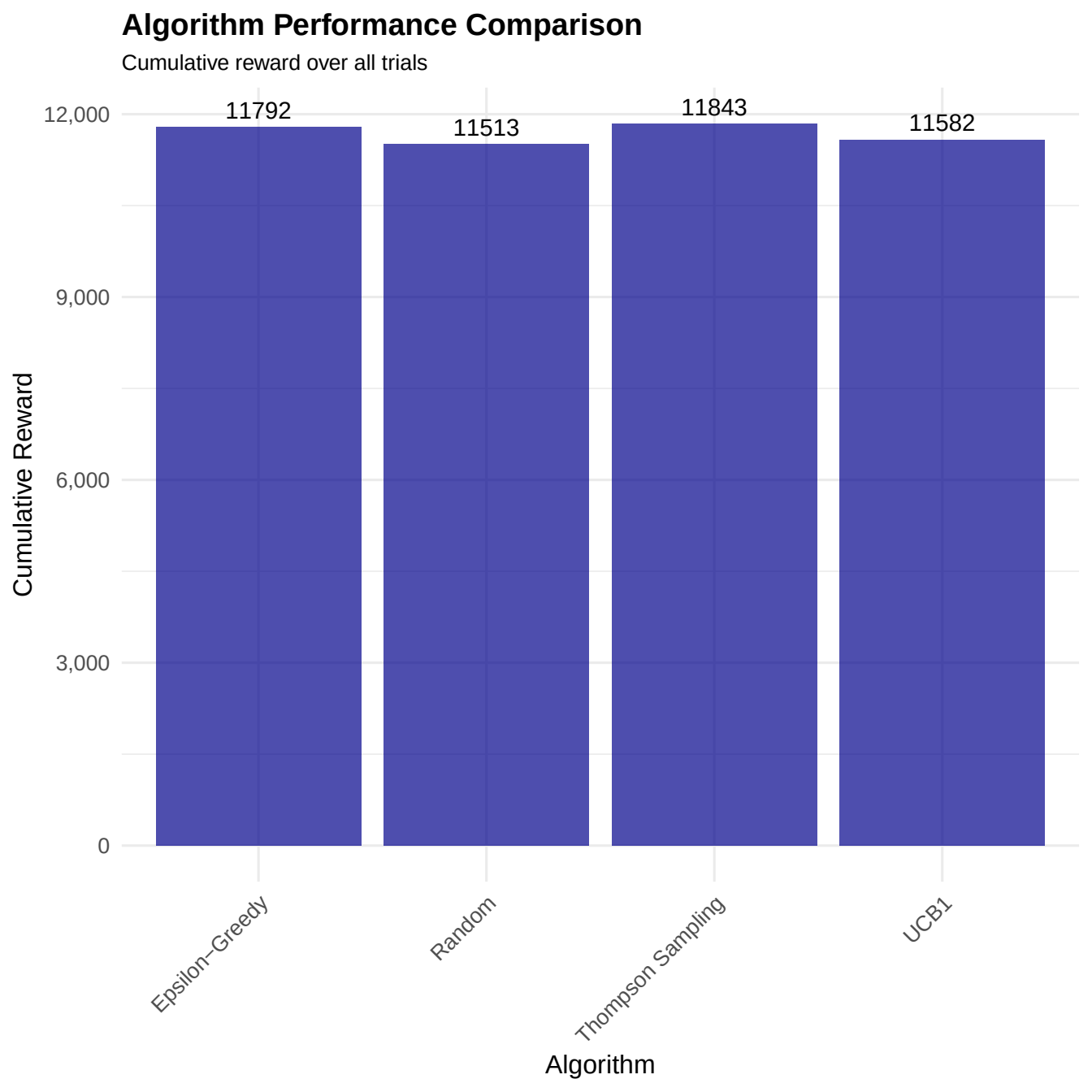
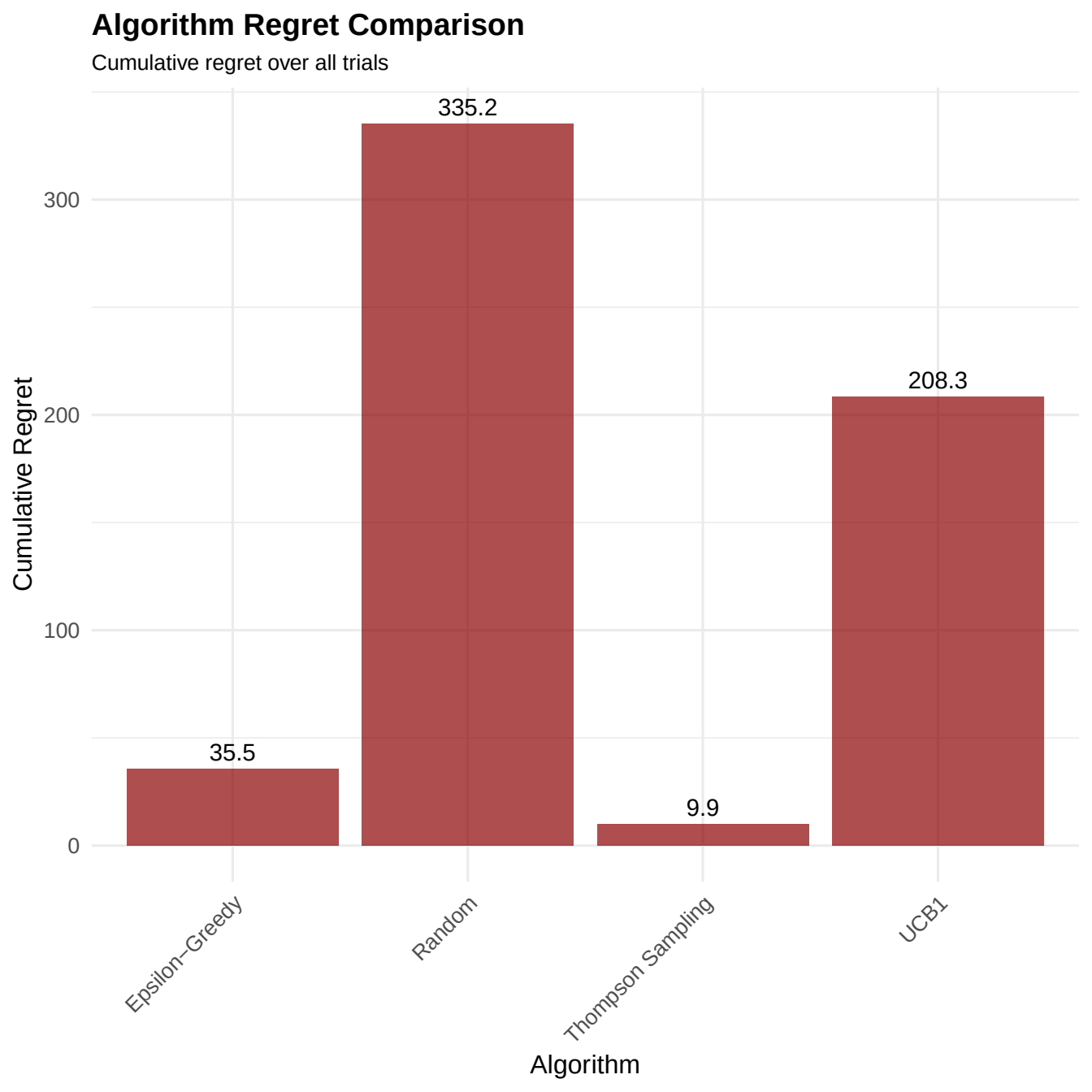

Bandit Arm Performance: Completion Rates
With 95% confidence intervals
88.6%
89.4%
93%
75%
50%
25%
0%
Completion Rate
price_first
rating_first
time_first
Recommendation Strategy (Arm)


Regret Analysis
Percentage regret vs. best arm
4.7%
3.9%
0%
4%
3%
2%
1%
0%
Regret (%)
price_first
rating_first
time_first
Recommendation Strategy (Arm)


Sample Size Distribution
Number of trials per arm
20,196
19,756
20,032
20,000
15,000
10,000
5,000
0
Number of Trials
price_first
rating_first
time_first
Recommendation Strategy (Arm)


Revenue Performance
Average revenue per trial
$2.00
$1.50
$1.00
$0.50
$0.00
Average Revenue per Trial ($)
$1.88
$1.88
$1.96
price_first
rating_first
time_first
Recommendation Strategy (Arm)


Algorithm Performance Comparison
Cumulative reward over all trials
Cumulative Reward
12,000
9,000
6,000
3,000
0
11792
11513
11843
11582
Epsilon−Greedy
Random
Thompson Sampling
UCB1
Algorithm


Algorithm Regret Comparison
Cumulative regret over all trials
Cumulative Regret
0
100
200
300
35.5
335.2
9.9
208.3
Epsilon−Greedy
Random
Thompson Sampling
UCB1
Algorithm

Bandit Arm Performance: Completion Rates
With 95% confidence intervals
Completion Rate
75%
50%
25%
0%
price_first
rating_first
time_first
Recommendation Strategy (Arm)
Regret Analysis
Percentage regret vs. best arm
Regret (%)
4%
3%
2%
1%
0%
3.9%
0%
price_first
rating_first
time_first
Recommendation Strategy (Arm)
Sample Size Distribution
Number of trials per arm
Number of Trials
20,000
15,000
10,000
5,000
0
price_first
rating_first
time_first
Recommendation Strategy (Arm)
Revenue Performance
Average revenue per trial
Average Revenue per Trial ($)
$2.00
$1.50
$1.00
$0.50
$0.00
price_first
rating_first
time_first
Recommendation Strategy (Arm)
Algorithm Performance Comparison
Cumulative reward over all trials
Cumulative Reward
12,000
9,000
6,000
3,000
0
Epsilon−Greedy
Random
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Algorithm
Algorithm Regret Comparison
Cumulative regret over all trials
Cumulative Regret
300
200
100
0
35.5
9.9
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Epsilon−Greedy
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